

PRD (Product Requirements Document)

Project Name

Donoris

A Smart Emergency Blood Donation & Donor Discovery Platform

1. Executive Summary

Donoris is a web-based platform designed to solve blood shortage challenges by connecting patients, hospitals, blood banks, and verified blood donors through a centralized and intelligent donor matching system.

The platform focuses on emergency blood requirements, especially rare blood groups, by providing real-time donor discovery, location-based matching, secure communication, donation tracking, and emergency broadcast notifications.

The primary objective is to reduce the time required to find eligible blood donors and increase blood donation participation across Pakistan.

2. Problem Statement

Pakistan faces significant challenges in blood availability due to:

- Fragmented blood bank databases
- Lack of centralized donor records
- Delayed donor discovery during emergencies
- Difficulty finding rare blood groups
- Privacy concerns among donors
- No efficient communication system between patients and donors
- Low donor retention and engagement

As a result, patients and hospitals often rely on social media posts and personal networks during critical situations.

3. Product Vision

To build Pakistan's largest intelligent blood donor ecosystem that can instantly connect blood seekers with eligible nearby donors while maintaining privacy, security, and donor motivation.

4. Goals & Objectives

Primary Goals

- Reduce donor search time from hours to minutes
- Create a nationwide donor database
- Increase repeat blood donations
- Improve accessibility of rare blood groups
- Enable emergency response mechanisms

Business Goals

- Partner with hospitals and blood banks
 - Build a trusted healthcare platform
 - Create a scalable infrastructure for nationwide adoption
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5. Target Users

1. Blood Donors

Citizens willing to donate blood voluntarily.

Capabilities

- Register profile
 - Manage availability
 - Receive donation requests
 - Track donation history
 - Earn badges and recognition
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2. Blood Seekers

Patients, family members, or caretakers seeking blood.

Capabilities

- Create blood requests
 - Search donors
 - Contact matched donors
 - Track request status
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3. Hospitals

Healthcare institutions requiring blood urgently.

Capabilities

- Submit emergency requests
 - Verify donations
 - Access donor recommendations
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4. Blood Banks

Partner organizations maintaining donor records.

Capabilities

- Sync donor databases
 - Verify donor eligibility
 - Update donation records
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5. System Administrators

Platform managers.

Capabilities

- Manage users
 - Verify organizations
 - Moderate requests
 - Monitor system health
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6. User Roles

Role	Permissions
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Role	Permissions
Donor	Manage profile, receive requests
Seeker	Search and request blood
Hospital	Create emergency requests
Blood Bank	Manage donor records
Admin	Full system access

7. Core Features

Feature 1: User Registration & Authentication

Description

Users can create accounts as donors or seekers.

Functional Requirements

- Email registration
- Phone registration
- OTP verification
- Password reset
- Login/logout
- Profile management

User Data

- Full Name
 - Gender
 - Age
 - Blood Group
 - City
 - Area
 - Contact Information
 - Medical Eligibility
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Feature 2: Donor Profile Management

Description

Each donor has a complete profile.

Profile Fields

- Blood Group
- Last Donation Date
- Availability Status
- Health Status
- City
- Area
- Donation Count
- Badge Level

Availability States

Active

Eligible for donation.

Temporarily Inactive

System automatically marks donor unavailable if:

- Donation occurred within last 90 days
 - Donor reports illness
 - Donor manually pauses availability
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Feature 3: Centralized Blood Donor Database

Description

Unified database combining:

- Registered users
- Partner blood banks
- Historical verified donor records

Requirements

- Duplicate detection
- Data verification
- Data synchronization

- Secure storage
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Feature 4: Smart Search & Matching Engine

Description

Automatically identifies the most relevant donors.

Matching Criteria

Primary Filters

- Blood Group Match
- Availability Status

Secondary Filters

- Same Area
- Same City
- Distance Radius

Advanced Filters

- Recent Activity
- Donation Frequency
- Response Rate

Search Flow

1. User enters required blood group.
 2. System identifies eligible donors.
 3. Location filtering is applied.
 4. Donors ranked by relevance.
 5. Top matches displayed.
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Feature 5: Emergency Blood Request System

Description

Allows creation of urgent blood requests.

Request Fields

- Patient Name
- Hospital Name
- Blood Group Required
- Quantity Required
- City

- Area
- Urgency Level
- Contact Person

Statuses

- Pending
 - Searching
 - Matched
 - Fulfilled
 - Expired
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Feature 6: Privacy-First Communication

Description

Protect donor privacy while enabling communication.

Requirements

- In-app chat
- Contact masking
- Request approval before contact
- No public phone numbers
- Female donor privacy protection

Benefits

- Safer interactions
 - Reduced harassment risk
 - Better trust
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Feature 7: Direct Communication System

Description

Enables communication between matched parties.

Features

- Real-time chat
- Request acceptance/rejection
- Message notifications
- Read receipts

Feature 8: Urgent Broadcast & Alert System

Description

Emergency notification mechanism for critical blood requests.

Trigger Conditions

When:

- No donor found
- Rare blood group requested
- Emergency request marked critical

Notification Channels

SMS

Send alerts to:

- Matching blood group
- Same city donors

Push Notifications

Instant mobile/browser alerts.

Email Alerts

Backup communication channel.

Broadcast Flow

1. Request created.
2. Matching fails.
3. Broadcast triggered.
4. Nearby donors notified.
5. Responses collected.

Feature 9: Donation Verification System

Description

Donation records must be verified.

Verification Methods

- Hospital confirmation
- Blood bank confirmation
- Admin verification

Stored Data

- Date
 - Location
 - Donation Type
 - Verification Source
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Feature 10: Gamification & Rewards

Description

Encourages repeat donations.

Badge System

Badge	Requirement
First Saver	1 Donation
Life Saver	5 Donations
Hero Donor	10 Donations
Community Champion	25 Donations
Blood Guardian	50 Donations

Additional Rewards

- Certificates
 - Leaderboards
 - Community Recognition
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8. Non-Functional Requirements

Performance

- Search results under 2 seconds
- Support 100,000+ users
- High availability architecture

Security

- Encrypted passwords
- JWT authentication
- HTTPS everywhere
- Role-based access control

Privacy

- GDPR-inspired principles
- Data encryption
- Consent-based communication

Scalability

- Cloud-ready architecture
 - Horizontal scaling support
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9. Admin Dashboard

Features

- User Management
 - Donor Verification
 - Blood Request Monitoring
 - Analytics Dashboard
 - SMS Campaign Management
 - Badge Moderation
 - Hospital Management
 - Blood Bank Management
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10. Analytics & Reporting

Metrics

- Total Donors
- Active Donors
- Donations Completed
- Requests Fulfilled
- Average Match Time
- City-wise Statistics

- Blood Group Distribution
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11. Suggested Technology Stack

Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS

Backend

- Node.js
- Express.js / NestJS

Database

- PostgreSQL

Cache

- Redis

Authentication

- JWT
- OTP Verification

Notifications

- Twilio SMS
- Firebase Cloud Messaging

Hosting

- AWS
 - DigitalOcean
 - Vercel
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12. MVP Scope (Phase 1)

Included

- User Registration

- Login
- Donor Profiles
- Blood Request Creation
- Donor Search
- Location-Based Matching
- Availability Toggle
- In-App Messaging
- Admin Dashboard

Excluded

- AI Predictions
 - Advanced Analytics
 - Hospital Integrations
 - Mobile Apps
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13. Future Enhancements

AI-Powered Matching

Predict donor responsiveness.

Mobile Applications

Android and iOS apps.

Real-Time Geolocation

Distance-based donor ranking.

Government Integration

National donor registry integration.

Blockchain Verification

Tamper-proof donation records.

Emergency Ambulance Integration

Direct emergency response workflows.

14. Success Metrics

KPI Targets

- 10,000 registered donors in first year
 - 80% emergency request fulfillment rate
 - Average donor match time under 5 minutes
 - 70% donor retention rate
 - 95% notification delivery success rate
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Conclusion

Donoris aims to become a nationwide blood donation ecosystem that intelligently connects blood seekers with eligible donors, improves emergency response times, protects donor privacy, and encourages continuous blood donation through recognition and community engagement.